

UNIT – 1
ASSESSMENT – 2
REPORT

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1. **Title:** Emergency Medicine Delivery Using A* Search Algorithm

Aim:

To analyze and compare Greedy Best-First Search and A* Search for AI-powered emergency drone navigation in dynamic environments.

Problem Statement:

To determine the safest and shortest route for delivering emergency medicines while considering changing weather conditions, blocked roads, and varying travel costs.

Algorithm Used:

- Greedy Best-First Search
- A* Search Algorithm

Software Requirements:

- Python 3.x
- Visual Studio Code / IDLE

Program Explanation:

The program represents locations as graph nodes and uses the A* Search algorithm to find the optimal route from the starting location

to the destination. The algorithm combines the actual travel cost and heuristic estimate to compute the most efficient path while minimizing overall cost.

Output:

Optimal Path: A -> B -> E -> G

Total Cost: 9

Result:

The comparison shows that A* Search is more suitable than Greedy Best-First Search for emergency drone navigation because it considers both the actual travel cost and heuristic estimate. This enables the drone to identify safer and more optimal routes, making A* highly effective in dynamic and uncertain disaster environments.

2. Title: Traffic Signal Optimization Using Genetic Algorithm

Aim:

To optimize traffic signal timings using AI techniques to reduce waiting time, fuel consumption, and traffic congestion.

Problem Statement:

To develop an intelligent optimization system that continuously adjusts traffic signals based on changing traffic conditions while achieving efficient traffic flow.

Algorithm Used:

- Hill Climbing
- Simulated Annealing

- Genetic Algorithm

Program Explanation:

The program demonstrates an optimization process by improving traffic signal timings. It evaluates different signal combinations and selects better solutions based on a fitness function representing traffic efficiency.

Output:

Optimized Signal Timings: [30, 40, 55, 45]

Fitness Value: 170

Result:

Genetic Algorithms are the most effective approach for smart traffic optimization because they explore multiple solutions simultaneously, adapt to changing traffic conditions, and avoid getting trapped in local optima. This makes them suitable for large-scale, real-time traffic management systems.

3. **Title:** Mars Rover Navigation Using an Online Search Agent

Aim:

To implement an online search approach for autonomous Mars rover navigation in an unknown environment.

Problem Statement:

To enable the rover to safely navigate, detect hazards, update its knowledge, and reach exploration targets without a complete map.

Algorithm Used:

- Online Search Agent

Program Explanation:

The program simulates a Mars rover exploring different terrain locations. The rover checks each location using sensor information, detects hazards such as rocks or craters, and chooses an alternative path whenever necessary. This demonstrates how an online search agent makes decisions based on real-time information.

Result:

The online search agent is the most suitable approach for Mars exploration because it allows the rover to make real-time decisions, adapt to unknown terrain, avoid hazards, and safely complete its mission.

4. Title: Exam Timetable Generation Using Constraint Satisfaction Problem (CSP)**Aim:**

To generate an exam timetable that satisfies multiple scheduling constraints using Backtracking Search and Forward Checking.

Problem Statement:

To assign exams to suitable time slots and halls without conflicts while satisfying faculty availability, hall capacity, subject order, and student requirements.

Algorithm Used:

- Constraint Satisfaction Problem (CSP)
- Backtracking Search
- Forward Checking

Program Explanation:

The program treats each subject as a variable and available time slots as the domain. Backtracking assigns slots while checking constraints. If a conflict occurs, it backtracks and tries another assignment. Forward Checking reduces future conflicts by eliminating invalid choices early.

Result:

Backtracking Search combined with Forward Checking is an efficient method for generating exam timetables. It satisfies multiple constraints, reduces conflicts, and improves scheduling performance, making it suitable for universities with a large number of courses and students.

5. Title: Game AI Using Minimax and Alpha-Beta Pruning**Aim:**

To develop an intelligent game-playing AI that makes optimal decisions efficiently using the Minimax algorithm with Alpha-Beta Pruning.

Problem Statement:

To enable an AI opponent to evaluate multiple game states, make optimal decisions within strict time limits, and efficiently handle large decision trees.

Algorithm Used:

- Minimax Algorithm
- Alpha-Beta Pruning

Program Explanation:

The program implements the Minimax algorithm to evaluate possible moves in a game. Alpha-Beta Pruning reduces the number of branches explored by eliminating moves that cannot improve the final decision. This speeds up the search while still producing the optimal move.

Result:

Minimax with Alpha-Beta Pruning is the most suitable approach for real-time strategy games because it provides optimal decisions while significantly reducing computation time. This enables the AI to respond quickly and compete effectively against human players.